

LECTURE 7

GEOSPATIAL RETRIEVAL, REASONING, AND AGENTS

Geospatial Representation Learning

Learning Outcomes

Lecture

- Explain how geospatial embeddings can support retrieval, question answering, and reasoning over spatial data.
- Describe geospatial AI agent components, including tools, data access, memory, reasoning, and evaluation.
- Compare manual geospatial analysis workflows with agent-assisted workflows.
- Critically assess the reliability, reproducibility, and limitations of AI agents for geospatial data analysis.

Lab

- Design a simple agent-assisted workflow for a geospatial analysis question.
- Connect an AI agent or scripted assistant to selected geospatial data tools or APIs.
- Evaluate the agent's outputs against a manually implemented baseline.
- Present limitations, failure cases, and possible improvements of the prototype workflow.



PRACTICAL 7

PROTOTYPE GEOSPATIAL AI AGENT

Geospatial Representation Learning